

Develop a Student Registration Portal using Java Servlets and Servlet Filters with request validation and preprocessing before form submission.

Index.html

```
<!DOCTYPE html>

<html>

  <head>

    <title>Student Portal</title>

  </head>

  <body>

    <h1>Registration Form</h1>

    <form action="register" method="post">

      Name: <input type="text" name="name">

      Course: <input type="text" name="course">

      Email: <input type="text" name="email">

      <button type="submit">Register</button>

    </form>

  </body>

</html>
```

ValidationFilter.java

```
package com.og.practical4c;

import jakarta.servlet.Filter;
import jakarta.servlet.FilterChain;
import jakarta.servlet.ServletException;
import jakarta.servlet.ServletRequest;
import jakarta.servlet.ServletResponse;
import jakarta.servlet.annotation.WebFilter;
import java.io.IOException;
```

```
@WebFilter("/register")
```

```
public class ValidationFilter implements Filter{
```

```
    @Override
```

```
    public void doFilter(ServletRequest req, ServletResponse resp, FilterChain chain)throws  
    IOException, ServletException{
```

```
        String name = req.getParameter("name");
```

```
        String course = req.getParameter("course");
```

```
        String email = req.getParameter("email");
```

```
        name.trim();
```

```
        course.trim();
```

```
        email.trim();
```

```
        if(name.isEmpty() || course.isEmpty() || email.isEmpty()){
```

```
            resp.getWriter().println("<h1>All Fields are mandatory.</h1>");
```

```
        }else{
```

```
            chain.doFilter(req, resp);
```

```
        }
```

```
    }
```

```
}
```

RegisterServlet.java

```
package com.og.practical4c;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.HttpServlet;
```

```
import jakarta.servlet.http.HttpServletRequest;
```

```
import jakarta.servlet.http.HttpServletResponse;
```

```
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/register")
public class RegisterServlet extends HttpServlet{

    protected void doPost(HttpServletRequest req, HttpServletResponse resp)throws
    IOException{

        String name = req.getParameter("name");
        String course = req.getParameter("course");
        String email = req.getParameter("email");

        PrintWriter out = resp.getWriter();
        out.println("<h1> Welcome "+name+" Registration Successfull.</h1>");
        out.println("<h1> Your Course is "+course+"</h1>");
        out.println("<h1> Your email is "+ email+"</h1>");
    }
}
```